

Player Performance Prediction (Group 1)

Team Github Repository

<https://github.com/yashs14282001/Group-1-Team-Project.git>

4. Pipeline Workflow

- Pipeline for the Player Performance Prediction Project provides the details on how the players' data would be cleansed, processed, trained, validated, and then deployed to develop a prediction application. Data collection phase for the dataset is underway, which will consist of various player performance data like match performance, batting /bowling records, and players' roles.

4.1 Data Pre-processing and Feature Engineering

- The first stage of the process will consist of examining the raw data set for gaps in data, duplicate data, the way data has been formatted, and the presence of data that is not needed. The numerical columns such as runs, wickets, strike rate, average, economy rate, number of games played and player ratings will be converted to the appropriate numerical data type. Categorical data columns such as player type, batting style, bowling style, team, and game type will also be converted to their respective numerical data types.
- An additional list of potentially helpful columns that are to be added to improve the accuracy of prediction includes:

Feature	Purpose
Total Runs	Measures overall batting contribution
Total Wickets	Measures overall bowling contribution
Batting Impact	Combines runs, average, strike rate, 50s, and 100s
Bowling Impact	Combines wickets, economy rate, and bowling average
Consistency Score	Measures performance across different matches or formats
Overall Performance Score	Target value used for prediction

- In order to ensure the quality of the data, validation procedures will be performed prior to feeding the data into the machine learning algorithm. As an illustration, the data about players' runs, wickets, strike rates, averages, and economy rates would be examined to identify any implausible numbers. Additionally, the same pre-processing procedures would be stored and used later when implementing the app.

4.2 Model Training and Improvement

- Once the preprocessed data is ready, it will be divided into training and test datasets, possibly at an 80:20 ratio. These models will predict either a numeric rating of performance or one of the defined categories (average, good, very good, excellent) of performance level.
- The models that will be used in this analysis include the following:

Model	Purpose
Random Forest	Baseline model for structured player statistics
Neural Network	Learns complex patterns in player performance
Hybrid Model	Combines batting and bowling impact predictions
Pre-trained Model	Uses transfer learning or pre-trained feature extraction

- Hyperparameter tuning will be performed to optimize. An example is for a Random Forest model, tuning will include numbers of trees as well as max depth.
- An example of hyperparameter tuning in a Neural Network model includes tuning the learning rate as well as the number of epochs, number of hidden layers, and batch size.
- Cross-validation will be done in training when applicable in order to prevent over-fitting of the model. It ensures that the accuracy of the model will be consistent throughout all aspects of the data and not just in one particular train-test separation. The most suitable model will be determined using its score and generalizability.

4.3 Model Evaluation and Metrics

- The model evaluation will be based solely on an unseen test dataset. All model evaluations for numeric performance prediction will be based upon the following numeric measures – Average Absolute Error (MAE), Root Mean Square Error (RMSE), and R-Squared (R2). For categorical performance predictions the following categorical measures will be used – Accuracy, Precision, Recall, F1 Score, and Confusion Matrix.
- A model comparison table will be included to clearly show the performance of each model. This will help justify why the final model was selected.

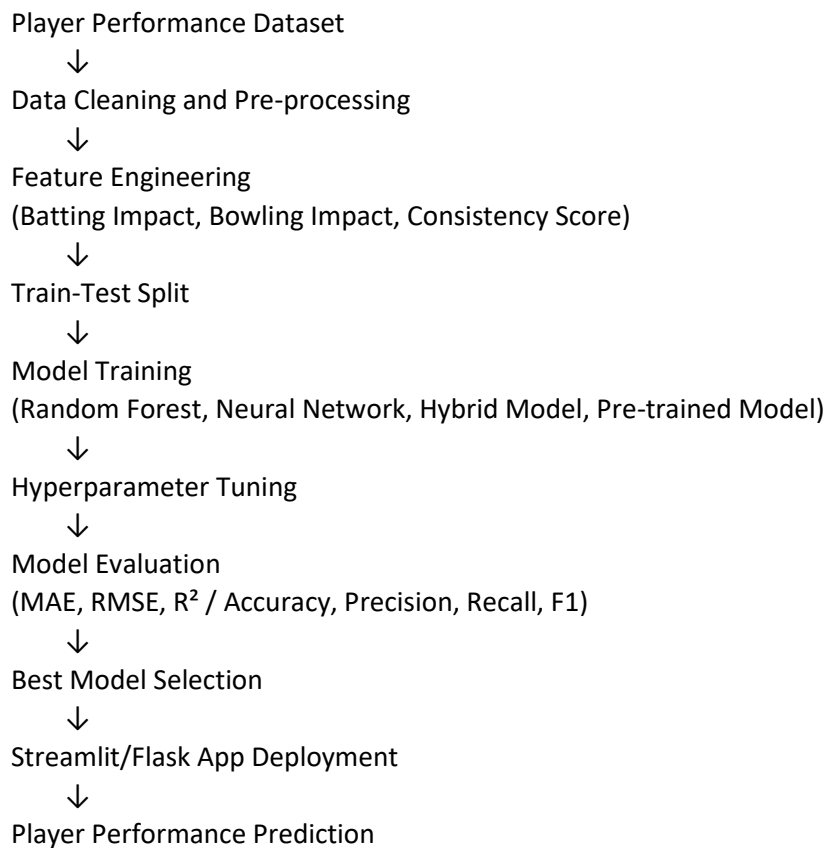
Model	Metric Result	Final Decision
Random Forest	To be calculated	Baseline comparison
Neural Network	To be calculated	Deep learning comparison
Hybrid Model	To be calculated	Custom combined approach
Pre-trained Model	To be calculated	Transfer learning comparison

- The model that demonstrates superior relative accuracy (or R-squared) and lower errors will be deemed to be the best performing model..

4.4 Application Deployment

- Once complete, the model that provides the best performance will be integrated into either a Streamlit or Flask application. The information provided to the user for input will consist of details regarding the players' position on their team, all the runs scored as a player, all wickets taken as a bowler, batting average, strike rate, economy rate, and type of match being played.
- The application will process the input and show the predicted result, such as:
 - Predicted Performance: Very Good or
 - Predicted Score: 8.4 / 10

Workflow Diagram



- The Player Performance Prediction process has been elaborate in the workflow presentation; from the preprocessing of the player data (e.g., removing empty fields/rows, getting rid of duplicate records, correcting any errors) through the creation of novel features based on the data set (e.g., batting impact, bowling impact, consistency score, runs scored, and wickets taken).
- To finally splitting the data into training and testing sets, building and optimizing multiple models via hyperparameter tuning, and determining which of the multiple models provided the best fit for the data by evaluating all models with appropriate metrics.
- In the end, the best fitting model will be implemented into either a Streamlit app or a Flask app, where the user can enter their player information and receive a prediction for that player's performance.